

ZINING HE

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Research Interests

Multimodal Foundation Models, Video Understanding and Generation, World Models, Efficient Large Language Models

Education

Shanghai Jiao Tong University

2024.09 – 2028.06

B.S. in Computer Science and Technology

Publications

SpecBridge: Natural Language to Lean Specifications | *NeurIPS 2026, accepted*

Third author

Industry Experience

Shanghai Chuangzhi Academy

Approx. 2 months

Algorithm Intern, Multimodal Large Models

Research Experience

SpecBridge: Natural Language to Lean Specifications

2026.03 – Present

Undergraduate Research Assistant

- Built a reconstruction-guided pipeline that translates natural-language requirements and fixed function signatures into formal Lean specifications.
- Designed NLFP planning and multi-layer validation with Python checks and Lean proof obligations, improving CLEVER results by 6.1/15.2/28.4 percentage points over few-shot + CoT baselines.

Adaptive Acceleration for MoE Large Language Models

2026.01 – Present

Project Member / SJTU Undergraduate Innovation Practice Program

- Designed adaptive expert allocation using gate scores, layer importance, attention, and token activations for MoE inference.
- Profiled Qwen1.5-MoE and DeepSeek-V2 with PyTorch hooks and MMLU scripts; reduced latency by 12% on Qwen1.5-MoE prefill and 5% on DeepSeek-V2 token generation.

Cross-Architecture Operator Differential Testing

2026.02 – Present

Undergraduate Research Assistant

- Developed a CUDA/CANN differential testing pipeline to compare neural-network operator semantics and support GPU/NPU reliability debugging.

Intent-Based Test Case Generation

2025.09 – 2025.12

Undergraduate Research Assistant

- Converted structured and semantic testing intents into executable tests for targeted coverage and bug finding.

Technical Skills

ML/LLM: PyTorch, large language models, MoE inference and evaluation

AI Systems: CUDA, Huawei CANN, GPU/NPU operator testing, inference profiling

Formal Methods: Lean, specification synthesis, proof- and test-based verification

Systems: Operating systems, computer architecture, Linux, Git

Languages: Python, C++, Rust, Java